

Day 2: Relativistic Kinematics I & II: Rapidity & Pseudorapidity in Detail

14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J, HEP-119

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Syllabus: 14-Day Hands-on Course Layout

Part I: Relativistic Kinematics

- Day 1: Foundations of Relativistic Kinematics & Experimental HEP
- **Day 2: Kinematics I & II: Rapidity & Pseudorapidity in Detail**
- Day 3: Kinematics III: Invariant Yield & Phase Space Jacobian
- Day 4: Kinematics IV: Coordinate Calculator implementations

Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity & p_T

Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles & m_T Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow (v_2)
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

Polar Angle to Pseudorapidity Mapping

Theoretical Definition (Sahoo Eq. 5.138)

At ultra-relativistic limits ($p \gg m_0$), the momentum rapidity y simplifies to a purely angular, geometric quantity called **Pseudorapidity** (η):

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right)$$

Where θ is the polar angle relative to the beam axis (z).

Physical Advantages of Pseudorapidity

- **No Particle ID (PID) needed:** Calculated solely from the emission angle θ of the trajectory in the tracker.
- **No magnetic field needed:** Does not require momentum information (only spatial tracking lines).
- **Symmetric / Odd behavior:** It is odd about $\theta = 90^\circ$ ($\eta = 0$). That is, $\eta(180^\circ - \theta) = -\eta(\theta)$, reflecting forward-backward symmetry.

Angular Detector Mapping: θ vs. η

Angle to η Reference (Sahoo Table 5.3)

Polar Angle θ	Pseudorapidity η	Detector Region
0°	∞	Forward Beam pipe
10°	2.44	Endcap region
30°	1.32	Forward transition
45°	0.88	Diagonal transition
90°	0.00	Central Barrel
135°	-0.88	Diagonal transition
170°	-2.44	Backward transition
180°	$-\infty$	Backward Beam pipe

Physical Trajectories (Sahoo Fig. 5.13)

- $\theta = 90^\circ$ ($\eta = 0$): Particles fly straight out transversely.
- $\theta \rightarrow 0^\circ$ ($\eta \rightarrow \infty$): Particles fly forward along the beam pipe.
- High $|\eta|$ values denote regions extremely close to the beam line (demanding high spatial resolution trackers).

Interactive Physics Simulator

Open in your browser:



[animations/05_pseudorapidity_detector](#)

(Interactive polar angle θ vs. pseudorapidity η detector mapping)

Coordinate Transformation: From (y, p_T) to (η, p_T)

Vector Decompositions (Sahoo Eq. 5.141 - 5.143)

By defining the pseudorapidity η in terms of longitudinal momentum p_z and total momentum magnitude $|\mathbf{p}|$:

$$e^\eta = \sqrt{\frac{|\mathbf{p}| + p_z}{|\mathbf{p}| - p_z}}, \quad e^{-\eta} = \sqrt{\frac{|\mathbf{p}| - p_z}{|\mathbf{p}| + p_z}}$$

We can solve for the momentum component variables in terms of (p_T, η) :

$$|\mathbf{p}| = p_T \cosh \eta, \quad p_z = p_T \sinh \eta$$

Which allows us to write the energy E and transverse mass m_T elegantly:

$$E = \sqrt{|\mathbf{p}|^2 + m_0^2} = \sqrt{p_T^2 \cosh^2 \eta + m_0^2}$$

The Phase-Space Jacobian Transformation

Rapidity to Pseudorapidity Mapping (Sahoo Eq. 5.147)

The distribution of particles as a function of pseudorapidity ($dN/d\eta$) is related to the rapidity distribution (dN/dy) via the phase-space Jacobian $J(y \rightarrow \eta) = dy/d\eta$:

$$\frac{dN}{d\eta dp_T} = J(y \rightarrow \eta) \frac{dN}{dy dp_T} = \sqrt{1 - \frac{m_0^2}{m_T^2 \cosh^2 y}} \frac{dN}{dy dp_T}$$

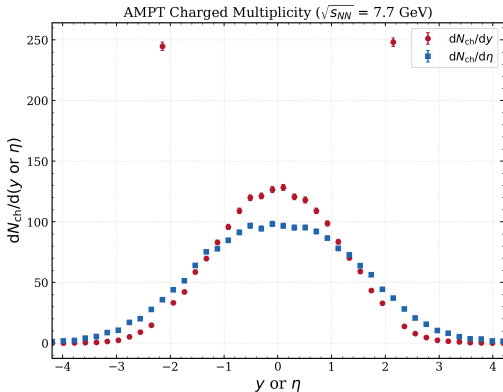
Where the Jacobian factor is exactly the particle's longitudinal velocity:

$$J(y \rightarrow \eta) = \frac{p}{E} = \beta$$

Physical Consequence of Jacobian

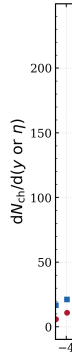
- ****Massless limit ($m_0 \rightarrow 0$):**** $J \rightarrow 1$, meaning $dN/d\eta \approx dN/dy$ (valid for photons or very high p_T hadrons).
- ****Massive limit ($m_0 > 0$):**** $J < 1$ at mid-rapidity ($y \approx 0$). This creates a local, non-physical suppression in $dN/d\eta$ even if the underlying physical dN/dy is flat.

AMPT Data Validation: dN/dy vs. $dN/d\eta$



Comparison of Yields (AMPT default)

- Plots show charged particle distributions in Au + Au collisions at $\sqrt{s_{NN}} = 7.7$ and 39 GeV.
- ****Rapidity (dN/dy)**** exhibits a higher peak at mid-rapidity and drops off faster.
- ****Pseudorapidity ($dN/d\eta$)**** is wider and has a slightly suppressed peak, caused directly by the integration over the Jacobian factor.

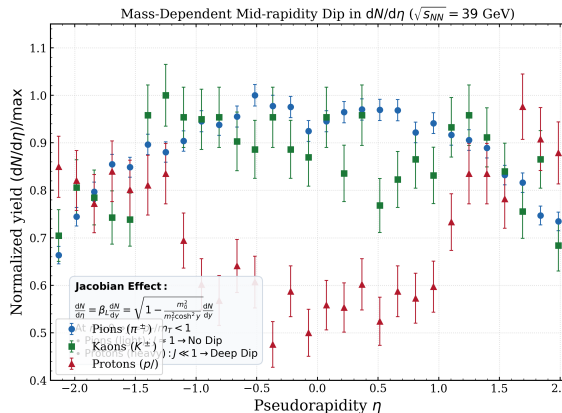


Interactive Jacobian Explorer

Open in your browser:

- [day2/day2_rapidity-jacobian-explorer.html](#)
(Interactive p_T , mass, and rapidity sweep dashboard)

Mass-Dependent Mid-rapidity Dip in $dN/d\eta$



Why do heavy particles dip at $\eta = 0$?

At mid-rapidity ($\eta = 0 \Rightarrow y = 0$), the Jacobian becomes:

$$J(0) = \frac{p_T}{m_T} = \frac{p_T}{\sqrt{p_T^2 + m_0^2}}$$

- **Pions ($\pi^\pm$, $m_0 \approx 0.14$ GeV):** Light mass. $J(0) \approx 1$. No dip is visible in the yield.
- **Kaons ($K^\pm$, $m_0 \approx 0.49$ GeV):** Medium mass. Shallow dip begins to emerge.
- **Protons ($p/\bar{p}$, $m_0 \approx 0.94$ GeV):** Heavy mass. $J(0) \ll 1$ at low p_T , creating a deep, pronounced mid-rapidity dip.

Takeaway

This dip is purely a coordinate mapping artifact. The physics (dN/dy) remains flat or peak-like.

Landau Hydrodynamics & Rapidity Widths

Landau Model Rapidity Distribution (Sahoo pg. 145-146)

In Landau's hydrodynamic model of relativistic expansion, the rapidity distribution of produced pions is described by a Gaussian:

$$\frac{dN}{dy} = \frac{N}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{y^2}{2\sigma_y^2}\right)$$

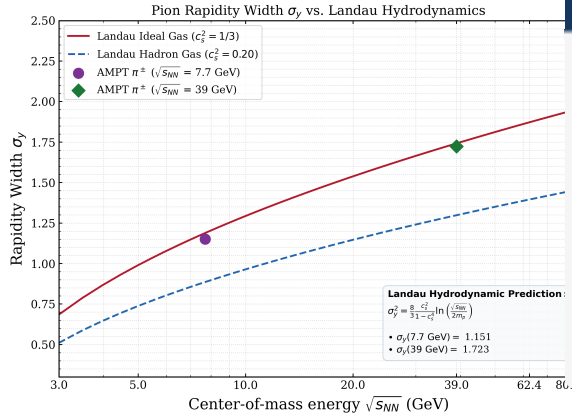
Where the width of the rapidity distribution σ_y relates to the center-of-mass energy and the speed of sound c_s in the medium:

$$\sigma_y^2 = \frac{8}{3} \frac{c_s^2}{1 - c_s^4} \ln\left(\frac{\sqrt{s_{NN}}}{2m_p}\right)$$

Landau Model Scaling Limits

- ****Ideal Gas limit ($c_s^2 = 1/3$):**** Prefactor becomes 1, so $\sigma_y = \sqrt{\ln(\sqrt{s_{NN}}/2m_p)}$.
- ****Hadron Gas limit ($c_s^2 \approx 0.20$):**** Premature freezing or lower speed of sound shrinks the rapidity width σ_y .

Equation of State: Softest Point of QGP



Speed of Sound c_s^2 Probe (Sahoo Fig. 5.16)

- By measuring the pion rapidity width σ_y in AMPT and experiments, we can extract the effective speed of sound $c_s^2 = \partial P / \partial \epsilon$.
- **The "Softest Point":**** In heavy-ion energy scans, c_s^2 exhibits a minimum (a "dip") around $E_{\text{beam}} \approx 30$ AGeV ($\sqrt{s_{NN}} \approx 7.7$ GeV).
- **Physics Interpretation:**** This minimum denotes the softest point in the Equation of State (EoS), which is a classic signature of the transition from hadronic matter to QGP (crossover or first-order phase transition).

Summary: Day 2 Hands-On Laboratory Exercises

Objective: Analytical & Data-Driven Rapidity Calculations

All exercises are framed in your Python files located in the `exercises/` and `day2/` folders:

- ❶ ****Problem 1:**** Calculate center-of-mass energy $\sqrt{s_{NN}}$ for RHIC Energy Scan values.
- ❷ ****Problem 2:**** Prove why collider configurations have a massive kinematic advantage over fixed-target configurations.
- ❸ ****Problem 3:**** Compute and verify Lorentz factors γ and velocities β for LHC Pb+Pb beams.
- ❹ ****Problem 4:**** Parse AMPT events, extract particle species (pions, kaons, protons), and plot the mass-dependent mid-rapidity dip in $dN/d\eta$.
- ❺ ****Problem 5:**** Extract the standard deviation (width σ_y) of pion rapidity and check compatibility with Landau's ideal gas limit vs. viscous gas models.